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Graded Assessment - Modular Exponentiation

Graded Quiz • 30 min

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Module 2 Discussion

Module Assessment

Quiz: Graded Assessment - Modular Exponentiation

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Graded Assessment - Modular Exponentiation

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Due Apr 7, 11:59 PM IST Attempts 3 every 8 hours

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1. What is $47 \cdot 69 \pmod{143}$

1 / 1 point

125

17

47

142

Correct

This was an example worked in the lesson video.

Try again

2. What is $15 \cdot 15 \pmod{14}$

1 / 1 point

13

15

-1

14

Correct

This also congruent to 1 (mod 14)

3. The value of totient(N) is equal to

1 / 1 point

The number of positive integers less than and relatively prime to N.

The number of positive integers less than N that share factors with it.

The number of prime numbers no larger than N.

The number of positive integers less than N that divide it.

Correct

This is the definition of Euler's totient function.

4. What is the totient(p) if p is prime?

1 / 1 point

p

(p-1)

(p-1)/p

p(p-1)/2

Correct

All numbers less than p are relatively prime to p.

5. What is the totient of 2717 (all prime factors are less than 20)?

1 / 1 point

2160

2717

40

1267

Correct

Prime factors of 2717 = {11,13,19}, tot(2717) = [tot(11)*tot(13)*tot(19)] = 10*12*18 = 2160

6. What is the totient of 3^4 ?

1 / 1 point

54

26

80

27

Correct

$\Phi(3^4) = \Phi(81) = 54$

7. Under what conditions is the totient function multiplicative?

1 / 1 point

Never

If the numbers don't have repeated prime factors.

If the two numbers are relatively prime.

Always

Correct

Correct. The multiplicative property does not hold universally.

8. If $x > y$, what is true regarding $\log(x)$ and $\log(y) \pmod N$ (if the base is greater than 1)?

1 / 1 point

Nothing reliable.

$\log(x) < x$ and $\log(y) < y$

$\log(x) > \log(y)$ (assuming both exist).

$\log(x)$ exists if $\log(y)$ exists.

Correct

Discrete logs have very little in way of reliable patterns.

9. If $\log(x) \pmod N$ exists to a particular base, then

1 / 1 point

It is possible that other numbers have the same log (to the same base).

It exists to any other base that is relatively prime to the first base.

It is unique.

It may or may not exist to other bases.

Correct

Discrete logs have very little in way of reliable patterns.

10. A primitive root

1 / 1 point

only exists if the modulus is prime.

is any base that is relatively prime to the modulus.

generates all numbers relatively prime to the modulus.

must be relatively prime to the modulus.

Correct

This is by definition.

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